

Mathematical Modelling — Tutorial 2

How these tutorials work. Each tutorial has pen-and-paper parts and a *computational task* that you carry out in whatever language you prefer (MATLAB is recommended; Python or Julia are equally fine). The point of the code is not the language, but that a classmate can regenerate your figure from your folder. Each week, you add a folder `weekNN/<username>/` to the course tutorials repository on GitHub, containing a half-page `report.md`, the figure(s) requested, the code that produced them, and a short `README` giving the command to regenerate them. You submit this as a *pull request* and review one classmate's in return (see the repository's `CONTRIBUTING.md`). Parts marked *Challenge* are optional, for the motivated or when time allows. Chapter and appendix references are to the course notes. Every `report.md` should also close with one sentence of *interpretation*: what does your result mean for the real system you modelled, and not merely as a piece of mathematics?

Tutorial 2 (Week 2 — Nondimensionalisation and the phase plane)

Notes: Chapter 3 on nondimensionalisation and dominant balance; Appendix on linear stability.

1. The undamped pendulum equation is $\ddot{\theta} + \omega^2 \sin \theta = 0$, where $\omega^2 = g/L$ is the squared natural frequency. Introduce the dimensionless time $\tau = \omega t$, and show that the equation becomes

$$\theta'' + \sin \theta = 0,$$

where primes denote differentiation with respect to τ , hence, explain why one may, without loss of generality, set $\omega = 1$.

2. Nondimensionalise the damped pendulum equation

$$mL\ddot{\theta} + k\dot{\theta} + mg \sin \theta = 0,$$

where $k \geq 0$ is the damping coefficient.

- (a) Write the resulting dimensionless equation as a first-order system.
- (b) Find all equilibria.
- (c) Using the Jacobian matrix and the trace–determinant classification (Appendix A), determine the type and stability of the downward equilibrium $(0, 0)$ for at least three different parameter regimes.
- (d) Classify the upright equilibrium $(\pi, 0)$, and compute its eigenvectors.

Computational task. Plot the phase portrait (direction field together with several representative trajectories) for each of the parameter regimes identified above. Confirm visually the classifications obtained in Question 2: the centre, stable spiral, and stable node at the downward equilibrium, together with the saddle at the upright equilibrium.

MATLAB users may find the ready-made `dfield/ppplane` tools convenient. In other languages, a direction-field plot (using `quiver` or `streamplot`) together with a numerical ODE solver is sufficient.

Challenge (optional). Suppose an external control torque

$$F = a\phi + b\dot{\phi},$$

is applied to the pendulum, where

$$\phi = \theta - \pi$$

measures the angular displacement from the upright equilibrium.

Starting from the dimensional equation

$$mL\ddot{\theta} + k\dot{\theta} + mg\sin\theta = F,$$

linearise about the upright equilibrium to obtain an equation of the form

$$\ddot{\phi} + p\dot{\phi} + q\phi = 0.$$

Express p and q in terms of a , b , k , m , L , and g . Hence determine the region of the (a, b) -plane for which the upright equilibrium is asymptotically stable. What does this tell you about the feedback required to balance the pendulum upright?

Repo deliverable. Submit:

- the phase-portrait figure(s); and
- a `report.md` listing all equilibria, their classifications, and a brief physical interpretation of each.

If you attempted the challenge, also include a labelled sketch of the stability region in the (a, b) -plane.